AUTONATEAI SCREEPS LEAGUE

First Invasion Defense

A local Screeps competition module

docs/competitions/01-first-invasion-defense.md

Summary

The learner triggers one real NPC invader wave against their local room and proves their colony can detect hostiles, fire towers, protect critical structures, and keep the economy alive.

Skill Target

- Primary: hostile detection and tower automation.
- Secondary: defensive population targets, repair triage, observation.
- Program track: Developer, Episode 7.
- Recommended episode: `docs/tutorials/07-defense.md`.

Prerequisites

- Local Screeps server is running.
- Sparring health check returns `ok: 1`.
- Learner owns one room with `Spawn1`.
- Room has at least one tower with energy.
- Learner code calls tower logic every tick.

Setup

Replace `W6N3` with the learner's room name.

```
curl http://localhost:21025/local/api/sparring/health
```

```
curl -X POST http://localhost:21025/local/api/sparring/wave \
  -H 'Content-Type: application/json' \
  -d '{"room": "W6N3"}'
```

Challenge

Survive the first invasion without manual combat commands. The learner may watch the room, but all attacks, repairs, and defender spawning must come from code.

Pressure

One engine-generated invader wave from the local sparring endpoint.

Win Condition

The learner passes when:

- The invader is destroyed.
- `Spawn1` survives.
- At least one worker survives or the colony automatically recovers its worker population.
- Tower behavior is visible in code, not manually controlled.

Scoring

Metric	Full Credit	Partial Credit
Outcome	Invader dies and economy continues.	Invader dies but recovery needs manual spawning.
Code behavior	Towers attack hostiles before repairs.	Towers work, but priority is unclear.
Debugging process	Learner records wave result, failure point, and one code change.	Learner describes result without evidence.

Replay

Run the same sparring wave command after restoring tower energy. If the room stops producing invaders, choose a room with an exit into a neutral room.

Reflection

- How did your code find the hostile?
- What did the tower do first?
- What structure or creep was most at risk?
- What is one automated recovery behavior you added after the wave?

Extensions

- Easier: allow one tower and no defender creep requirement.
- Harder: require tower refueling while the wave is active.
- Team variant: one learner writes tower logic, one writes population recovery.